

Shaping the Internet of Energy with Edge Computing



Edge Computing: the Next Step in the Digital Journey

Accelerate the Digital Journey with Edge Computing

Edge computing is increasingly impacting many industries, including the energy sector. Edge-enabled, high-voltage products will build the foundation for the Internet of Energy (IoE). At the same time, edge computing at substation level is a useful add-on service which does not compete with the cloud at enterprise level.

The digital substation of the future needs more agility around high-voltage applications to manage increasing grid complexity. Furthermore, the latest system updates should always be available. This can be ensured by edge-enabled systems with built-in update managers. Increased system resiliency and continuous operations in case of any faults or damages are secured with backup and recovery features that are core elements of edge systems. Edge computing can be integrated seamlessly and secured into the customers' information technology (IT) landscape without any media breach between IT and operation technology (OT).

New use cases with higher response time requirements will be possible with edge computing, combined with digital twins and artificial intelligence. Hybrid approaches (combining cloud and edge systems) will appear, with a clear shift of OT applications to the edge layer and fleet and grid views in the cloud.

The Next Energy Evolution is Around the Corner

Multi-sourced energy generation and supply are characterized by fluctuating power flows and higher grid complexity. Grid operators must deal with these challenges. Digitally enabled assets can help manage the increasing complexity of transmission grids with built-in intelligence at product, system and substation levels for more flexibility and efficiency.

The new capabilities go far beyond the current monitoring, control and automation systems. Furthermore, for an interoperable, digital and decentralized network, it is necessary to interconnect high-voltage transmission products in the substation. These include transformers, gas-insulated switchgears and circuit breakers. Operational data from intelligent assets can be analyzed in real time in order to provide recommendations for improving performance. This kind of integration and operation builds the Internet of Energy (IoE) for the grids of the future.

The Internet of Things (IoT) plays a significant role in the digital transformation of organizations and industries worldwide. The same is true for the ongoing digitalization of the energy sector. This is the reason we call the evolving, digitalized system Internet of Energy.

A growing number of sensors and smart devices can be connected to the internet. However, in many cases, it is simply not feasible to connect thousands or even millions of IoT devices directly to the cloud. This is where edge computing comes in.

The IoT edge layer is typically located between field assets and the IoT cloud platform. The main features of an edge solution focus on connecting the assets, processing and managing the data locally on the substation level, moving data to the cloud platform for further data analytics, learning and integrating with backend services as well as rolling out firmware and software updates.



Increasing Importance of Edge Computing in Various Industries

Important Advantages of Edge Computing

By facilitating data processing at the edge of a network – close to the source – edge computing helps overcome some typical challenges of IoT deployments.

Flexibility and developer centricity: A containerized edge architecture makes it possible to apply fit-for-purpose languages, scale applications selectively, reuse cloud applications without rewriting code and access advanced edge functions.

Focus on business value: Edge computing enables cloud or custom logic on the device giving enterprises more value from diverse edge assets. It also helps users to process and act on IoT data as well as manage devices from the cloud. Device manufacturers can tap new revenue streams with connected products and expedite hardware and software development.

Heterogeneous edge landscapes: For many business-critical use cases, it is not sufficient to connect single devices of the same sort. Often, a mixture of different device classes – from microcontrollers built into devices up to industrial PCs – must be connected to collect and process business-critical data. Edge computing is a means to orchestrate such heterogeneous landscapes.

End-to-end IoT: By connecting previously unconnected devices, like microcontrollers, IoT edge is the key to realizing end-to-end IoT use cases and rolling out software updates via gateways – even down to small micro-controllers.

Network latency: Network latency is a significant issue for safety-critical systems, like a self-driving car or a factory control system. The difference between a response time of 100ms and 1ms can be life threatening. Edge computing allows for reduced latency due to local processing of key computer decisions.

Data privacy and security: Reducing the attack vector of an IoT application helps to create a more secure system. Edge computing reduces the number of devices connected to the internet. Local data filtering reduces the amount of sensitive data being transmitted.

Network load reduction: By processing data locally, edge computing can significantly reduce network bandwidth requirements. This is particularly important in cases where

Advantages of edge computing:

- Flexibility and developer centricity
- Focus on business value
- Heterogeneous edge landscapes
- End-to-end IoT
- Network latency
- Data privacy and security
- Network load reduction
- Computational efficiency
- Reduced cloud costs
- Autonomy

bottlenecks might occur due to unreliable and constrained network connectivity.

Computational efficiency: It is often more efficient to perform data processing and analytics on smaller data sets and with longer inter-arrival times. For example, more efficient analyses can be performed on a smaller data set (such as smaller geo-specific areas) and data that arrives with a longer time interval (in the order of seconds instead of milliseconds).

Reduced cloud costs: Local storage and processing do not only reduce network load. They also avoid sending unnecessary data to the cloud, consequently decreasing the costs associated with cloud storage and processing.

Autonomy: For certain use cases, it is critical that the local application can continue to run in disconnected mode. Edge computing allows for local execution, so a system can continue to run autonomously, even with no network connectivity.

Furthermore, edge opens up new ways of deploying value-add applications based on analytics and digital twin technology offline on the substation level.

High-frequency data captured from the assets is preprocessed, stored and analyzed near the source of data generation. As a result, fast application response times and reduced network bandwidth requirements are possible.

Increasing Relevance of Edge Computing

Given the benefits of edge computing it is no surprise it is gaining relevance and traction in many industries. The following examples represent just a very small selection of use cases from different domains. The examples provide insight into how edge computing solves some of the challenges of IoT deployments.

Industrial IoT

There are many industrial IoT use cases requiring quick response times enabled by edge computing.

For safety-critical systems, a millisecond response time is essential to avoid physical injury and machine damage. For instance, when a human enters a restricted area in a factory, machines must stop immediately. In this case, the ability to make fast, deterministic decisions at the edge is critical for human safety and machine preservation in industrial IoT applications.

For predictive maintenance and condition monitoring applications, data processing can be performed locally on an edge device. This reduces the amount of sensitive data sent outside a factory and improves the overall privacy and security of the application.

Smart Home

The smart home environment, covering several different applications and the need to connect previously unconnected domains and devices, is a major use case for edge computing.

Central smart home gateways supporting different protocols act as hubs to connect a multitude of devices to the cloud. This minimizes security threats and maximizes privacy and data protection. Edge computing enables local processing of data that users want to keep private and only transmits anonymized alarm signals if notifications are needed.

It allows for central roll-out of software and firmware updates to smart home devices, helping to keep up with the latest security patches, new functionalities and value-adding business models.

Retail

Supermarkets and retail stores can benefit from IoT applications to improve the efficiency of their buildings and customer experience.

Regulations, for example, require the temperature and humidity levels of cooling systems to be monitored and logged to prove food has been safely stored. This case is known as hazard analysis and critical control points (HACCP). Usually, the systems are manually checked once a day and this takes up a lot of time. In addition, many stores

are closed Sundays and public holidays, so if a cooling system fails, no one will notice, and the result is spoiled food.

An edge computing solution enables local temperature, humidity and air pressure monitoring without involving the cloud. Thus, cloud costs can be reduced and any change in conditions can trigger notifications locally. Consequently, corrective actions can be taken before any food spoils.

Automotive

Edge computing is well suited for automotive and other mobility use cases like autonomous driving, and remote monitoring and diagnostics of automobiles. An edge gateway can run on a vehicle's telemetry control unit to enable the following functionalities:

Communication: Edge software can act as the communication client to enable the exchange of control information for device management and telemetry data.

Applications: Edge software can run user applications on the automobile, such as an emergency call service (eCall) that provides precise locational details in the event of an accident. Another example is a diagnostics application that collects logging information from different components within the vehicle to identify any problems.

Updates: Edge software can be used to coordinate software updates of the vehicle's electronic control units, head unit, or navigation map material.

Transmission Industry

Edge computing is well suited for high-voltage transmission applications, including real-time overload, gas and oil leakage prediction, supported by artificial intelligence and machine learning. It also includes autonomous cooling control, remote monitoring and diagnostics of the assets including transformers and switchgears. In particular, the edge layer can run closer to the high-voltage equipment to enable more functionality, including:

Communication: Wide variety of communication protocols, such as OPC UA, ModBus TCP, IEC 61850, MQTT are available to add the necessary integration flexibility to the user's IT environment.

Applications: Edge software runs applications such as digital twin-based features like simulation and prediction, residual lifetime calculation and AI-based apps to predict oil leaks or partial discharges.

Updates: Edge software can coordinate substation equipment software updates, or parameter updates or new applications.

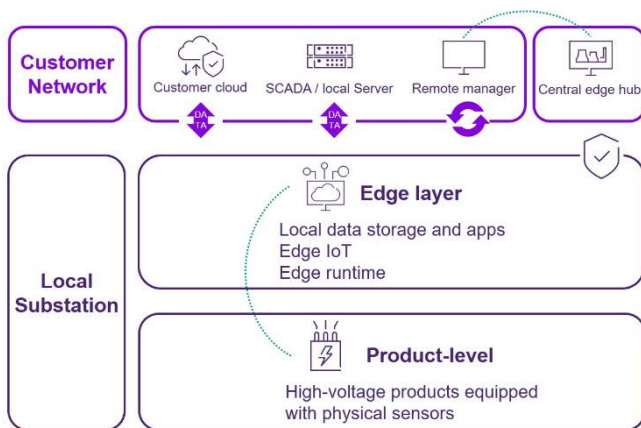
Edge Computing in the Substation

Edge Computing is Reshaping the IoT Topology in the Substation

Substation equipment is traditionally partially disconnected or connected to the local PC in the substation or to a SCADA system.

The core of the edge topology in the substation is the edge layer. The edge layer consists of four main elements: central edge hub, remote manager, edge IoT incl. runtime and apps.

The central edge hub serves as an artifact repository where applications and operation frameworks are stored and managed. Secondly, the remote manager is the customer interface for monitoring and updating the installed software. The edge IoT is the local hardware device that collects and processes the data as well as operates edge runtime. Finally, the apps are the most important parts of the edge layer and increase transparency, productivity and reliability by analyzing data provided by real and virtual sensors. The edge topology goes far beyond the classic substation control and monitoring and augments it with powerful cloud features.



Better Operational Performance with Edge Computing in the Substation

Keeping substation data offline and only sending information to the IT system outside the substation is a key requirement in a power transmission setting. Edge fulfills this requirement without compromising the seamless, secure and easy integration into the existing customer IT landscape.

An important advantage for the grid operator is the ease with which updates and data backups for the OT systems can be provided. Traditionally, these systems are either not up-to-date or have to be updated manually. The manual update process is costly, requiring knowledgeable experts to perform the task. Users will benefit from the latest software, firmware, cybersecurity updates and most recent applications. Automating this IT process leads to IT cost reductions.

Furthermore, edge opens up new opportunities for value-add applications based on analytics, AI and offline digital twin technology at the substation level. High-frequency data captured from the assets is pre-processed, stored and analyzed near the source of generation. As a result, fast application response times and reduced network bandwidth requirements are possible.

Conclusion

The increasing relevance of edge computing will also impact digital high-voltage products, systems and substations in the power transmission environment.

The Internet of Energy is based on intelligence and flexible assets and systems. Greater asset flexibility is important to cope with new grid dynamics and complexity.

Edge computing is the core of the next generation of digitally enabled high-voltage products, systems and solutions. It brings the intelligence and scalability of the cloud directly to the substation assets – with ease, top performance and no need for data to leave the local substation network. This provides extra flexibility to make informed and timely decisions. Edge products combine the strengths of local and high-performance data processing with the benefits offered by the cloud, including app-based data analytics. It also includes data processing and storage concepts with a centralized device and update management.

The interplay between edge and cloud combines the best of both worlds by wisely sharing computing resources according to application requirements. Edge computing covers local, highly responsive applications and the cloud offers a global fleet view that learns from the fleet and interconnected assets.

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